

SaniGuard Concept Note

Safely Managed Sanitation Hackathon

Team Name:

SaniGuard Alliance

Project Title:

SaniGuard: Climate-Resilient Smart Sanitation System

1. Executive Summary

SaniGuard is a low-cost climate-resilient sanitation system designed to protect vulnerable communities in Northern Ghana from sanitation failures during floods and droughts. The solution combines raised flood-resistant sanitation units, youth-led waste collection services, decentralized treatment hubs, and simple digital monitoring tools to safely manage sanitation waste across the entire sanitation chain.

Many communities in Northern Ghana experience seasonal flooding, water scarcity, poor waste disposal systems, and limited sanitation infrastructure. During floods, toilets overflow and contaminate water sources, increasing the spread of diseases such as cholera and diarrhea, especially among children. During droughts, water-dependent sanitation systems often fail.

SaniGuard addresses these challenges by introducing elevated low-water or dry sanitation units that remain functional during climate shocks. Waste is safely collected by trained youth sanitation teams using tricycles and sealed containers, then transported to community treatment hubs where it is processed into compost or reusable resources.

The solution also includes a low-cost digital reporting system using WhatsApp, SMS, and QR codes to enable communities to report sanitation failures, flood damage, and maintenance needs in real time.

SaniGuard's core innovation lies in integrating climate adaptation, sanitation management, youth employment, and community monitoring into one scalable and affordable sanitation ecosystem suitable for vulnerable communities in Northern Ghana.

2. Technical Design & Climate Resilience

Flood/Drought Adaptation

SaniGuard is specifically designed to function during climate shocks common in Northern Ghana.

Flood Adaptation Features

- Raised toilet structures above flood levels
- Sealed containment chambers to prevent leakage
- Elevated handwashing stations
- Emergency mobile sanitation response teams
- Leak-proof waste transport systems

These measures reduce contamination of groundwater and floodwater during heavy rains.

Drought Adaptation Features

- Low-water sanitation systems
- Dry sanitation options for water-scarce communities
- Water storage tanks for hygiene stations
- Water-saving hygiene education campaigns

This ensures sanitation systems continue operating even during prolonged dry periods.

Infrastructure

Hardware Components

- Raised sanitation units
- Ventilated improved pit systems or dry toilets
- Handwashing stations
- Waste collection tricycles
- Sealed waste tanks
- Solar drying beds
- Composting units
- Protective equipment for operators

Software & Digital Components

- WhatsApp reporting channels
- SMS alert system
- QR-coded sanitation monitoring
- Google Forms/data collection tools

- Simple dashboard for tracking maintenance and sanitation incidents

Materials

The system prioritizes affordable and locally available materials such as:

- concrete blocks,
- metal roofing sheets,
- plastic storage tanks,
- PVC piping,
- locally fabricated tricycles.

This reduces costs and improves local maintenance capacity.

3. Data Strategy & Operational Logic

Missing Data

Northern Ghana has limited groundwater, drainage, and sanitation mapping data. SaniGuard addresses this challenge through community-based data generation.

The project will:

- conduct community sanitation assessments,
- use GPS-enabled mobile phones for mapping,
- collect flood-risk observations from residents,
- identify sanitation hotspots through field surveys.

Over time, this information will support local sanitation planning and climate adaptation.

Maintenance Strategy

Maintenance will be managed through trained local youth sanitation teams operating under the SaniGuard model.

Their responsibilities include:

- routine inspections,
- waste collection,
- sanitation repairs,

- emergency response after floods,
- monitoring sanitation conditions.

To ensure sustainability:

- communities contribute small service fees,
- local assemblies and NGOs may support operations,
- compost and reusable products can generate additional income.

Following climate shocks, emergency response teams will rapidly inspect and restore damaged sanitation units.

4. Social Equity & Inclusion

SaniGuard prioritizes vulnerable groups including:

- children,
- women and girls,
- low-income households,
- persons with disabilities,
- flood-prone rural communities.

The sanitation units are designed to be:

- child-safe,
- accessible,
- affordable,
- suitable for schools and clinics.

To improve affordability:

- locally sourced materials reduce construction costs,
- phased payment/community contribution models can be used,
- partnerships with District Assemblies, NGOs, and donors can subsidize implementation.

The youth employment model also promotes inclusion by creating local green jobs and technical skills opportunities for unemployed youth.

5. Implementation Roadmap & Risks

Phase	Key Activities
Community Assessment	Identify pilot community, assess sanitation and flood risks
Design & Planning	Finalize sanitation design and operational structure
Prototype Development	Build sanitation prototype and monitoring system
Youth Training	Train sanitation operators and community monitors
Pilot Deployment	Install pilot units and begin operations
Monitoring & Evaluation	Track usage, maintenance, and climate resilience performance

3-Day Bootcamp (Hackathon)

During the 3-day bootcamp, the SaniGuard team will complete:

- a prototype sanitation system design,
- a sanitation chain workflow model,
- a digital reporting mockup,
- a pilot community implementation plan,
- a visual pitch deck and system architecture diagram.

The team will also develop:

- estimated pilot costs,
 - climate resilience strategy,
 - operational workflow,
 - impact measurement indicators.
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Risk Mitigation

Biggest Risk

The biggest risk is financial sustainability and long-term maintenance after deployment.

Mitigation Plan

SaniGuard will address this through:

- youth-led sanitation service operations,
- low operational costs,
- partnerships with local government and NGOs,
- community ownership models,
- revenue generation from compost and sanitation services.

The project also minimizes technical risks by using simple, locally maintainable technologies instead of expensive or highly complex systems.

Conclusion

SaniGuard aims to build safer, healthier, and climate-resilient communities by ensuring sanitation systems continue functioning during floods and droughts. By combining sanitation infrastructure, youth employment, waste treatment, and digital monitoring, the solution offers a scalable and sustainable model for Northern Ghana.